$$\left\{\text{Only 1-cycles} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \omega & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \omega^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \omega^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},\right.$$

$$\text{Only 1- and 2- cycles} = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{Only 1- and 2- cycles} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & \omega^2 & 0 & 0 & 0 \\ 0 & 0 & -\omega^3 & \omega & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 & \omega & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega^3 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 \\ 0 & 0 & \omega & 0 & 0 & 0 & 0 & -\omega^3 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{Only 1- and 2- cycles} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & -\omega^3 & 0 & 0 & 0 & \omega^2 & \omega & 0 & 0 \\ 0 & 0 & -\omega^3 & \omega & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & -\omega^3 & 0 & 0 & 0 & \omega & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & \omega & 0 & 0 & 0 & -\omega^3 & \omega^2 & 0 & 0 \\ 0 & \omega^2 & 0 & 0 & 0 & \omega & -\omega^3 & 0 & 0 \\ 0 & 0 & \omega & \omega^2 & 0 & 0 & 0 & -\omega^3 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{Only 1- cycles and all 2's} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & \omega & 0 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & \omega^2 & 0 \\ 0 & 0 & \omega^2 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & \omega & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & \omega & -\omega^3 & 0 & 0 \\ 0 & 0 & -\omega^3 & \omega^2 & 0 & 0 & 0 & -\omega^3 & 0 \\ 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{All 1's and all 2's} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\text{All 1's and 2's and a 0-loop} = \begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \\ 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 \\ 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 & -\omega^3 & 0 \\ 3 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & -\omega^3 & -\omega^3 & 0 & 0 \\ 0 & -\omega^3 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 \\ 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & -\omega^3 & 0 \\ 2 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \end{pmatrix},$$

$$\left.\text{All except 0-loops} = \begin{pmatrix} 2 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 3 \\ 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\omega^3 & 0 \\ 3 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 2 \end{pmatrix}\right\}$$